# Chapter 7

## Engineering Technology: Bridging Theory and Practice

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An engineering technologist is a specialist dedicated to apply basic engineering principles and technical skills (hands-on) to develop and implement existing technology. Therefore, engineering technology programs tend to put greater emphasis on hands-on laboratory skills.

### 7.1. Computer Engineering Technology

Computer engineering technologists work within the industry in supportive roles. The pathway to a computer engineering technology job can be shorter than it is with a traditional engineering degree -- you may break into the field with either an associate's or a bachelor's. If you do opt for a four-year-degree, you'll get a slightly different education than if you opted for computer engineering.

Computer engineering technology is more focused on pragmatics than theory. You won't need as much engineering, and you may not need as much calculus either. What can you actually do with your education? Plenty. Purdue University calls

engineering technologists practitioners, noting that some engineering programs grew out of the 1950's and 1960's push to be the most technologically adept nation. Excellence in this new frontier required conceptualizers -- people who could create new knowledge. Only a portion of today's engineering jobs, though, actually require conceptualizers; the other require practitioners. Graduates of high caliber engineering technology programs do take on engineering roles, though they can't compete for all the positions they could if they had an engineering degree and an engineering license.

The ABET accrediting agency notes that associate level computer engineering technologists are prepared to take on tasks like building, testing, maintaining and operating; baccalaureate candidates can also develop and implement systems. The computer engineering technology curriculum will give you a foundation in computer engineering concepts like electrical circuits, semiconductors, and embedded systems.

What are computer engineering technologists doing on the job? Options include quality assurance, computer integrated manufacturing, technical documentation, and computer repairs. Schools also report that their graduates have been hired as computer systems engineers, computer test engineers, and software engineers. Prospective students should be aware that many engineering positions do favor those with bachelor’s degrees.

## *Computer Engineering Technology Salary and Career Outlook*

What can you expect to earn if you pursue computer engineering technology – and how in demand will your skill set be? This depends on many factors including your educational level and the number of years you spend on the job. The Bureau of Labor Statistics provides some insight on what you can aspire to:

Jobs for network and computer systems administrators were expected to grow 23% between 2008 and 2018. An average salary of $82,250 was reported in 2025 with 1-4 years of experience.

Computer support specialists saw an average of $53,100 during the same time frame and same amount of experience. It should be noted that computer support technician positions only require a two-year associate degree. Computer systems analysts, who generally do have a bachelor’s degree listed on their resume, saw a mean annual wage of $81,250.

As with other careers, it takes experience to reach these income levels. Some graduates pursue certifications to validate their competency in areas like testing or quality assurance.

## *Selecting a Program in Computer Technology*

Computer technology programs can be accredited by ABET's Technology Accreditation Commission. There are separate standards for associate and bachelor's level programs.

Both programs will teach you the basics of electrical circuits, operating systems, microcomputers, electronics, and local area networks; they will also teach you how to apply these concepts to building, testing, operating, and maintaining systems. Bachelor's programs will introduce you to project management – an integral part of the software engineering life cycle. Systems analysis is also taught at the bachelor's level.

There is a good deal of variety in approach. Some programs combine computer and electrical engineering. Some include field work as a degree requirement. Others give you the option of completing an internship for elective credit. Indiana State University Reports that some students opt for minors like business to increase their employment options. (Software Engineer Insider, 2019)

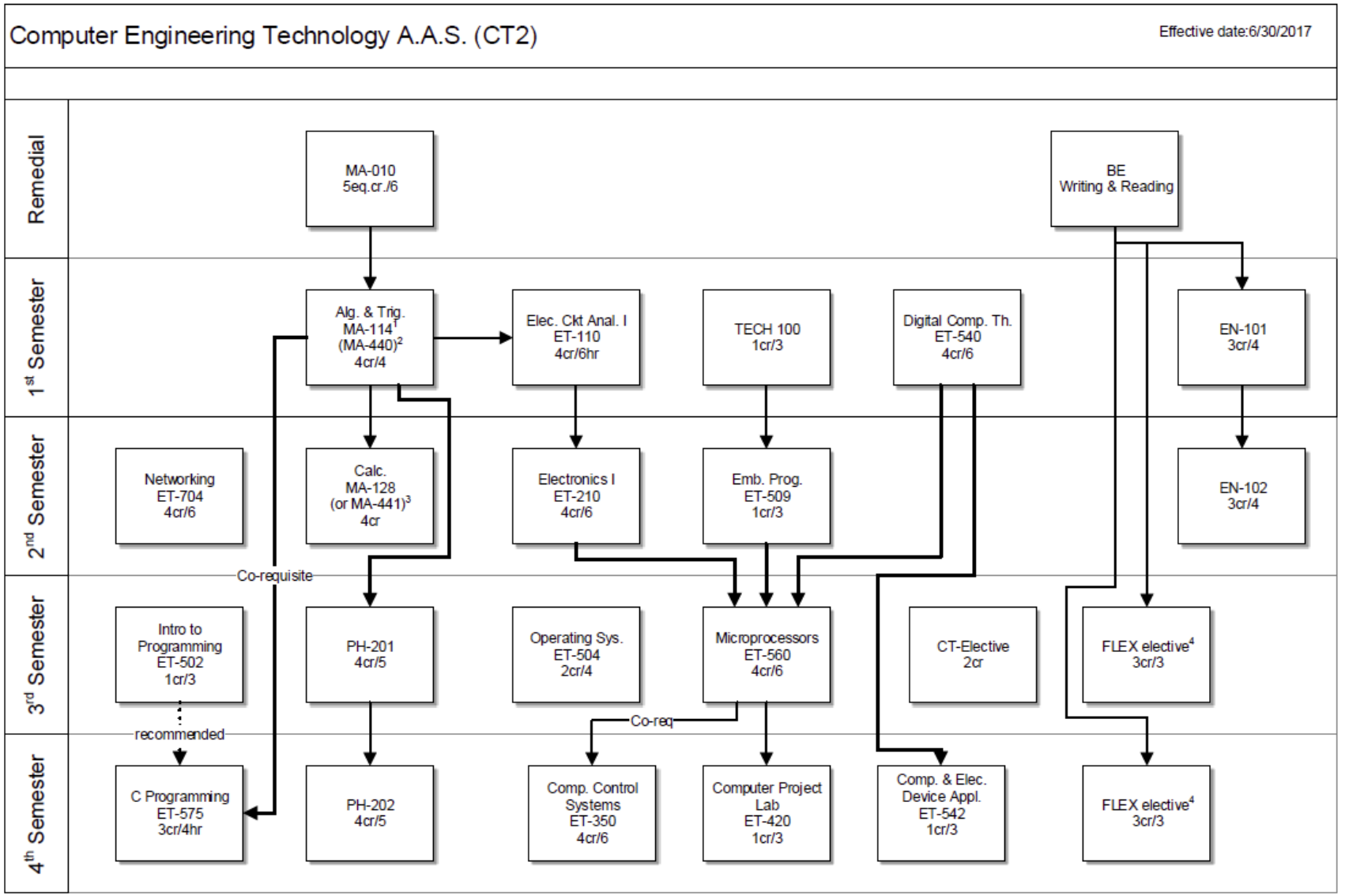
## *Computer Engineering Technology program in QCC*

The Associate in Applied Science (A.A.S.) degree program in Computer Engineering Technology combines electronics and digital theory as they relate to computers. Providing a solid foundation in both hardware and software, the program prepares students to accept positions in computer design and repair, embedded microprocessors and microcontrollers, appliance and control applications, and in networking applications. Students in the program learn in a hands-on, laboratory-based environment.

The Computer Engineering Technology program at Queensborough is one of the largest and most respected programs of its kind in the United States and is accredited by the Engineering Technology Accreditation Commission of ABET (ETAC of ABET). Additional information on ETAC of ABET can be found at http://www.abet.org.

Computer engineering technology is among the largest and most rapidly developing technical fields. Students in the program learn in a real-world environment that includes projects throughout the curriculum. They receive a solid foundation of coursework in digital and analog electronic systems, solid state systems, micro-computer and microcontroller systems, industrial process control and automation, computer programming and applications, local area networks and telecommunications. A wide variety of elective courses are also offered in specialty areas.

Computer Engineering Technology program graduates have an excellent record in finding employment and enjoy relatively high starting salaries. They are employed in such diverse positions as computer technician, LAN technician, computer technician, research technician, technical sales representative, technical writer, and field or customer engineer. New York area companies employing our graduates include: Verizon, Hewlett Packard, Consolidated Edison, IBM, MCS/Canon, Aeroflex, Kepco Power Supplies, CitiCorp, MTA, LIRR, US Postal Service, Keyspan, Lucent, Northrop/Grumman and Underwriters Laboratories.




[1] Taking both MA-119 & MA-121is equivalent to MA-114
[2] Must take MA-114 or equivalent before MA-440
[3] Must take MA-440 before MA-441
[4] Requires passing BE reading only. Elective chosen from FLEX IIA, B, D or E (ONLY HI 100 series, ANTH, CRIM101 OR 102, ECON, PLSC, PSYCH, OR SOCY)

*CT Electives:*
*ET-140 AC Circuits (highly recommended for transfer) 3cr/6 hr*
*ET-220 Electronics II 4cr/6hr*
*ET-230 Telecommunications 4cr/6hr*
*ET-232 Mobile communications 3hr/4cr*
*ET-375 Robotics 4cr/6hr*
*ET-481 PC Tech (A+) 2cr/4hr*

*ET-506 Python Programming 3cr/4hr*
*ET-580 Obj. Oriented Prog. 3cr/4*
*ET-705 Net Fund. II 4cr/5hr*
*ET-710 Web Tech. I 4cr/6hr,*
*ET-712 JavaScript Prog. 3cr/4hr,*
*ET-725 Security 3cr/3hr*
*ET-991, 992, 993 Coop Ed.*

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10/16/2018

## 7.2. Electronics Engineering Technology

Electronic devices of all kinds have become a permanent fixture of daily life in the 21st century; in fact, the Consumer Technology Association forecasts significant growth in a wide range of electronics sub-industries, including everything from smart home products to virtual reality systems. Electronic engineering technologists (hereafter referred to as EETs, for short) are some of the front-line workers who make this abundance possible.

### *How To Become an EET*

EETs learn a lot on the job, but finding an employer willing to train one entirely from scratch is a rare occurrence. Virtually all EETs need to learn the basics in a formal school curriculum. Classes on fundamental concepts like circuitry and control systems form the core of these programs, allowing students to handle sophisticated electrical systems safely and effectively.

As electronics have evolved in complexity over the years, classes on programming and the use of diagnostic tools have also become commonplace. Virtually all of these classes feature a practical element on top of the expected academic study. Many EET students take on work placements and internships too, giving them a chance to apply some of the things they've learned in a real-world setting.

It should be noted that EETs frequently use their elective courses to develop different specialties during their schooling depending on the larger industry in which they wish to work. Some choose to focus more on programming and internal systems, while others feel they are better suited to working with wiring and physical components. All trained EETs have a basic level of competency in all of these areas, but those with more education in particular path are able to set themselves apart from the rest.

### *What Does an EET Do?*

As you might guess from the well-rounded skill set they acquire in school, EETs can perform a wide range of duties in the workplace. Their typical primary tasks involves testing, maintenance, and repairs in both the design and implementation phases of electronics projects.

With diligence and proper documentation processes, this reveals any flaws or weaknesses in the design and the trigger events that caused them to fail; in this way,

EETs ensure that engineers have the data they need to address any problems with the current iteration and refine it into the best possible product.

EETs may also work under engineers more directly by collaborating with them during the production process, performing basic programming and design duties as dictated to help bring a product concept to life. They may be responsible for tracking down and source the parts the engineers need to create working design prototypes. Sometimes, though, they aren't able to find the exact type, size or shape of part that is called for; that's why they are also expected to fabricate custom versions from raw metal using tools like lathes and thread mills.

Finally, some EETs are occasionally called on to do things like write reports and speak to customers about any issues or possible improvements they would like to see in the products they buy. Strong communication skills are vital in this role, as they must be able to convey the information they discover both clearly and concisely for it to be of use to the overall project.

### *Can EETs Further their Careers after Working in the Field?*

It is important to re-emphasize the fact that EETs and engineers are two very different kinds of workers. There is nothing preventing EETs from upgrading their skills in the future. Electronics engineering technology training is an excellent starting point to build on if you have any interest in a being a fully-fledged electronics engineer. If you enjoy the material and the work it qualifies you to do, you can look into upgrading your credentials later. (ECPI University, 2019)

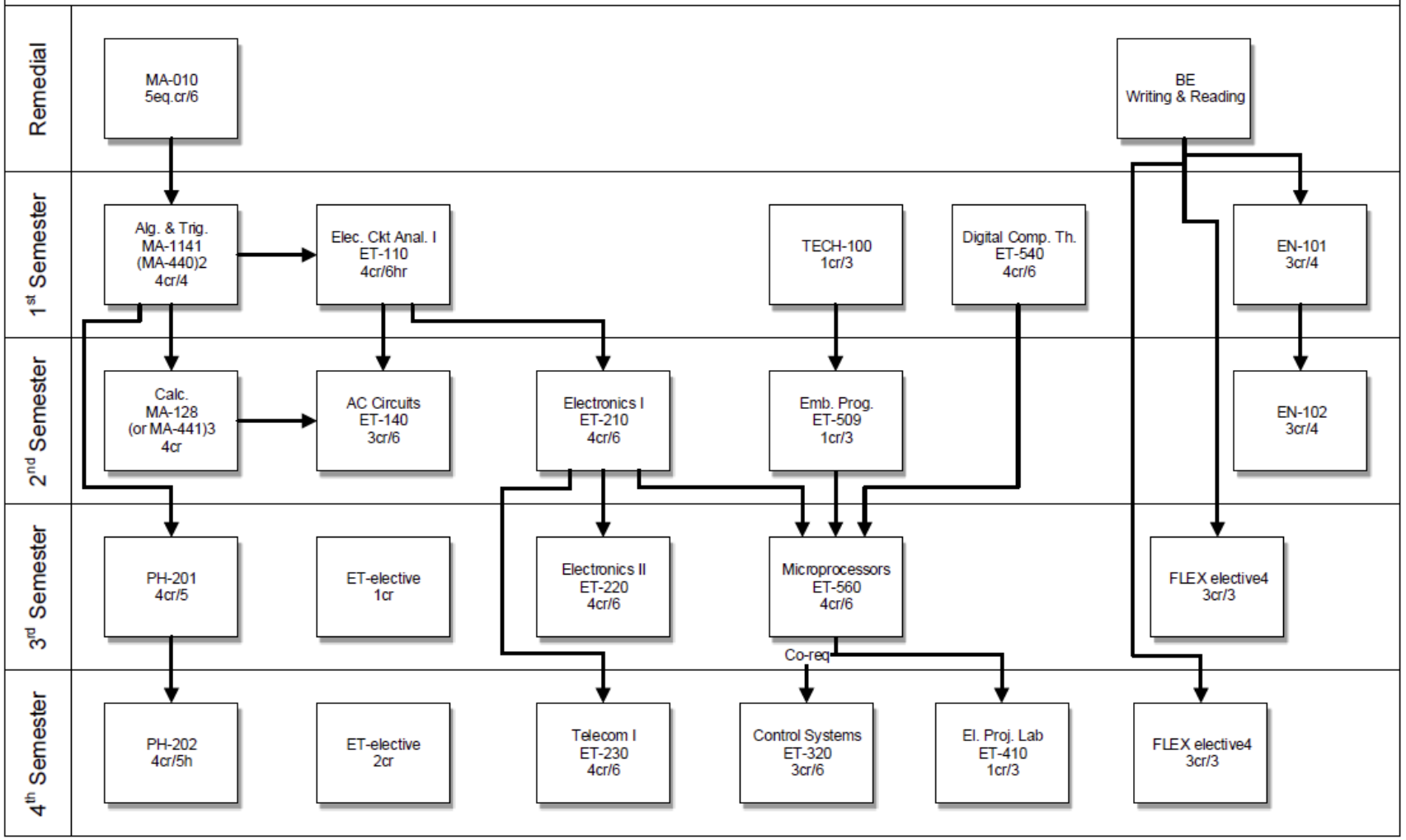
## 7.3. Electronic Engineering Technology program in QCC

The Associate of Applied Science (A.A.S.) degree program in Electronic Engineering Technology is a highly structured, laboratory-based curriculum that combines hands-on prac- tice with appropriate basic electrical and electronic theory. The program is applications-oriented and is designed to prepare well-rounded technicians who will compete and succeed in one or more of the fields related to electronic engineering technology.

The Electronic Engineering Technology program at Queensborough Community College is one of the largest and most respected programs of its kind in the United States and is accred- ited by the Engineering Technology Accreditation Commission of ABET (ETAC of ABET). Additional information on ETAC of ABET can be found at http://www.abet.org.

# Electronic Engineering Technology A.A.S. (ET2)

Effective date: 6/30/2017




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[4] Requires passing BE reading only. Elective chosen from FLEX IIA, B, D or E (ONLY HI 100 series, ANTH, CRIM101 OR 102, ECON, PLSC, PSYCH, OR SOCY)

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*ET Electives:*
*ET-305 Transients & Elec. Trans 2cr/3hr,*
*ET-360 Electronics & Automation for the Home 4cr/6hr*
*ET-375 Robotics 4cr/6hr*
*ET-481 PC Tech (A+) 2cr/4hr*
*ET-502 Intro to Programming 1cr/3hr*
*ET-504 Operating Systems Deployment 2cr/4hr*
*ET-505 C++ 4cr/6hr*
*ET-574 Python 4hr/3cr*
*ET-575 C Programming 3cr/3hr*
*ET-704 Net. Fund. I 4cr/6hr*
*ET-725 Computer Security 3hr/3cr*
*ET-991, 992, 993 Coop Ed.*

4/3/17

## 7.4. Mechanical Engineering Technology

A Mechanical Engineering Technology (MET) program at a two-year college (typically an associate degree program) is a hands-on, application-focused program that prepares students to work in mechanical design, manufacturing, testing, and maintenance. Unlike theoretical engineering programs, MET emphasizes practical skills using modern tools, machines, and software.

Curriculum Focus:

- Applied Mechanics (statics, dynamics, strength of materials)
- Mechanical Design (CAD/drafting, machine elements)
- Manufacturing Processes (CNC, welding, fabrication)
- Thermodynamics & Fluid Mechanics (basic level)
- Mathematics (through algebra, trigonometry, and sometimes calculus)
- Computer-Aided Design (CAD) and simulation tools
- Instrumentation & Testing
- Technical Communication and teamwork

Learning Environment:

- Heavy emphasis on labs and project-based learning
- Use of industry-standard tools and software (e.g., SolidWorks, AutoCAD, CNC machines)

Career Paths After Graduation:

Graduates typically qualify for technician or technologist roles, such as:

- Mechanical Engineering Technician
- CAD Designer/Drafter
- Manufacturing or Quality Technician
- Product Development Assistant
- Maintenance or Field Technician

Some may transfer to a 4-year program to earn a Bachelor's in Mechanical Engineering Technology or a related field.

## *Mechanical Engineering Technology program in QCC*

***Mechanical Engineering Technology*** is the application engineering principles to create useful machinery or products. Topics include Computer Aided Design (CAD) software, computer-aided manufacturing (CAM), and computer numerically controlled (CNC) machining.

The department on Engineering Technology in QCC offers an Associate in Applied Science Degree in Mechanical Engineering Technology. The MET major is accredited by the Engineering Technology Accreditation Commission of ABET (www.abet.org).

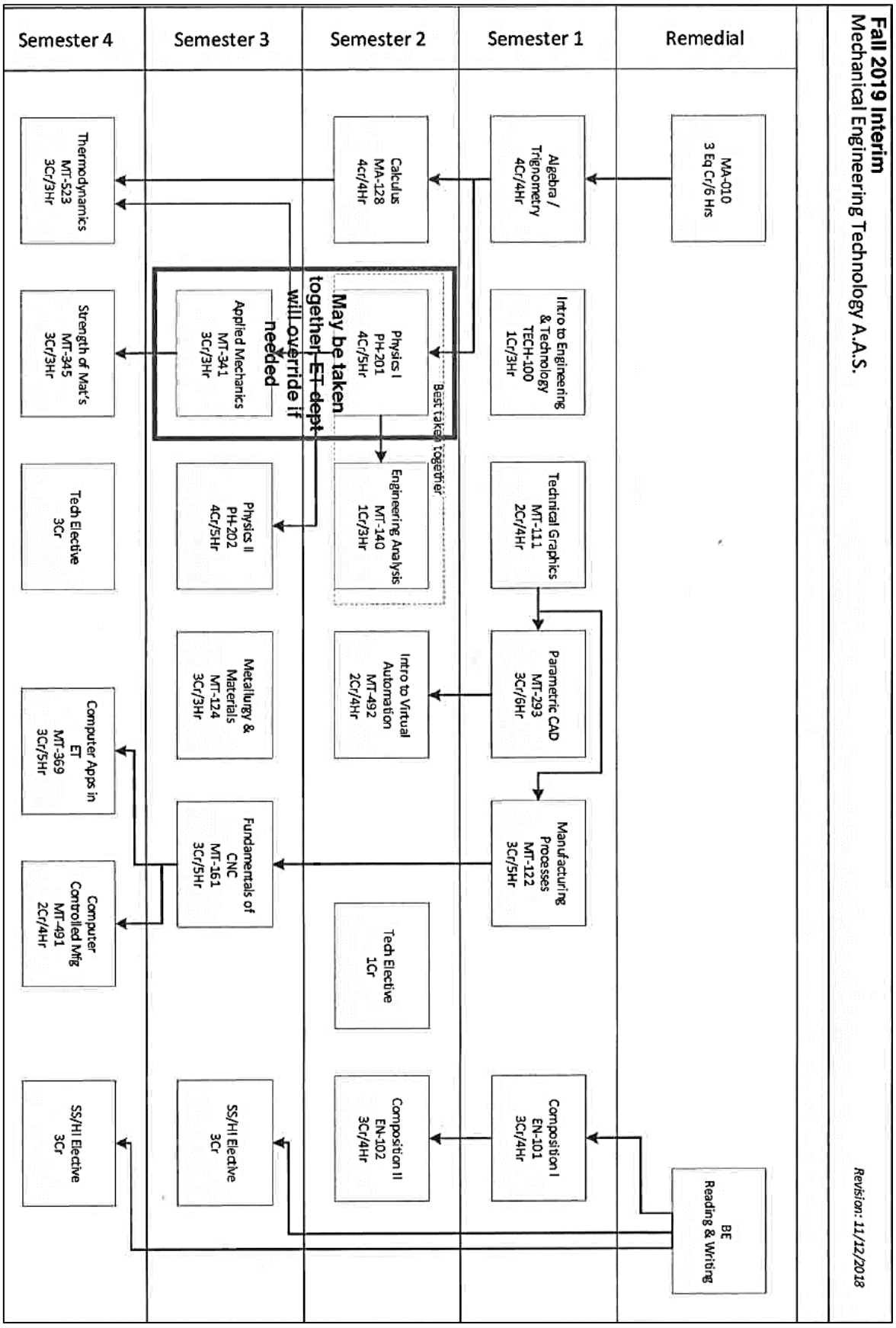
Graduates are competent technicians prepared with problem-solving skills, a strong technical foundation, and the mathematics and science education needed to enter the field of computer technology. Graduates may also transfer into the junior year of a bachelor's program in engineering technology.

The A.A.S. degree in Mechanical Engineering Technology is equivalent to the first two years of study for the Bachelor of Engineering Technology program of New York City College of Technology. Graduates of the Mechanical Engineering Technology program are eligible to apply for advanced standing to New York City College of Technology, with no loss of credit. Other colleges offer graduates of this program transfer credit into Bachelor of Mechanical Engineering degree programs.

Students can also transfer credits to the state college of Farmingdale State College. FSC offers three curriculum programs in mechanical engineering:

- **Bachelor of Science in Mechanical Engineering Technology**
- **Bachelor of Science in Manufacturing Engineering Technology**
- **Bachelor of Science in Facility Management Technology - Industrial Technology**

Contact FSC for more information about the number of credits that will be transferred from QCC to FSC.


Fall 2019 Interim
Mechanical Engineering Technology A.A.S.
Revision: 11/12/2018
Remedial
Semester 1
Semester 2
Semester 3
Semester 4
MA-010
3 Eq Cr/6 Hrs
Algebra / Trigonometry
4Cr/4Hr
Calculus
MA-128
4Cr/4Hr
Thermodynamics
MT-523
3Cr/3Hr
Intro to Engineering & Technology
TECH-100
1Cr/3Hr
Physics I
PH-201
4Cr/5Hr
Applied Mechanics
MT-341
3Cr/3Hr
Strength of Mat's
MT-345
3Cr/3Hr
Technical Graphics
MT-111
2Cr/4Hr
Engineering Analysis
MT-140
1Cr/3Hr
Physics II
PH-202
4Cr/5Hr
Tech Elective
3Cr
Parametric CAD
MT-293
3Cr/6Hr
Intro to Virtual Automation
MT-492
2Cr/4Hr
Metallurgy & Materials
MT-124
3Cr/3Hr
Computer Apps in ET
MT-369
3Cr/5Hr
Manufacturing Processes
MT-122
3Cr/5Hr
Tech Elective
1Cr
Fundamentals of CNC
MT-161
3Cr/5Hr
Computer Controlled Mfg
MT-491
2Cr/4Hr
Composition I
EN-101
3Cr/4Hr
Composition II
EN-102
3Cr/4Hr
SS/HI Elective
3Cr
SS/HI Elective
3Cr
BE
Reading & Writing
Best taken together
May be taken together ET dept will override if needed

# References

*ECPI University*. (2019). Retrieved from EET Degree Description: What is Electronics Engineering Technology?: https://www.ecpi.edu/blog/eet-degree-description-what-is-electronics-engineering-technology

*Software Engineer Insider*. (2019). Retrieved from A Career in Computer Engineering Technology: What is it?: https://www.softwareengineerinsider.com/careers/computer-engineering-technology.html